THOMAS LENZ

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SUMMARY

Third-year Ph.D. student in Industrial Engineering and Human-Computer Interaction studying how people work with automated systems. Runs human-subjects experiments end to end — protocol, apparatus, data collection, and analysis — using physiological, behavioral, and self-report measures to evaluate interfaces for workload, trust, and situation awareness. Three peer-reviewed publications and six conference presentations. Seeking a 2027 summer internship in UX or human factors research.

EDUCATION

Ph.D., Industrial Engineering and Human-Computer Interaction Expected Spring 2029
Iowa State University, Ames, IA

M.S., Industrial Engineering Expected Spring 2027
Iowa State University, Ames, IA

B.I.Sy.E., Industrial Engineering, Minor in Computer Science Spring 2024
University of Minnesota–Twin Cities, Minneapolis, MN

Award: Harold and Shirley Reihman Graduate Scholar, Iowa State University

SKILLS

Research Methods	Human-subjects experimentation, user and usability evaluation, design of experiments, statistical modeling, machine learning, survey design
Measurement	Eye tracking, physiological signal collection, behavioral coding, self-report instruments
Programming & Analysis	Python, R, MATLAB, Excel, AMPL, Simio, Qualtrics, vibe coding
Simulation & Prototyping	Unreal Engine, Unity, CARLA, VR, CAD

RESEARCH EXPERIENCE

Graduate Research Assistant — HIA Lab Fall 2024 – Present
Iowa State University, Ames, IA

- Design and evaluate human-vehicle interfaces in simulated automated driving environments, measuring operator workload, trust, and situation awareness.
- Currently collecting data from 32 participants on an Unreal Engine 4 driving simulator to test how display compellingness affects attention during partially automated driving.
- Developing an open-source simulator in CARLA with a Python API to assess situation awareness during remote assistance of autonomous vehicles.
- Conducted a systematic review of human factors interventions that improve remote-assistant situation awareness; presented at the IISE Annual Conference.
- Mentor undergraduate researchers on experiment preparation, simulator development, and data analysis.

Project Mentor — NSF Research Experience for Undergraduates Summer 2025
Iowa State University, Ames, IA

- Led three undergraduates through a full research cycle in ten weeks: research question, experiment plan, simulator build, and participant testing.
- Built a CARLA-based simulator to test how queue design affects the performance of remote operators supervising automated vehicles.
- Produced a prototype teleoperation simulator, a pilot study, a draft paper, and a poster presentation.

Undergraduate Research Assistant — HumanFIRST Lab Fall 2023 – Fall 2024
University of Minnesota, Minneapolis, MN

- Collected and analyzed field data on pedestrian crossing behavior and driver yielding behavior.

- Evaluated the efficacy of environmental interventions on pedestrian safety; contributed to a published assessment of two pedestrian safety inventories.

Teaching and Industry Experience

Teaching Assistant — IE 2710 Applied Ergonomics and Work Design Spring 2026; Spring 2027
Iowa State University, Ames, IA

- Developed the cognitive ergonomics curriculum and delivered two lectures to the undergraduate course.
- Ran labs and in-class activities and held regular office hours.

Product Engineering Intern Summer 2023
MVE, New Prague, MN

- Developed a testing protocol to assess the durability of cryogenic dewars.
- Designed experiments that minimized testing cost while maximizing statistical power.
- Evaluated the usability of dewar products.

Jr. Engineer Trainee Fall 2021 – Fall 2023
St. Anthony Falls Laboratory, Minneapolis, MN

- Supported faculty and graduate experimental setup; fabricated metal, plastic, and wood apparatuses.

Publications

Lenz, T., Bahng, E. J., Park, S., Liu, J., & Weng, B. (2026). A survey of future teachers suggests cautious openness to VR and humanoid robot use. *Proceedings of the Human Factors and Ergonomics Society Annual Meeting.* doi:10.1177/10711813261475235

Lenz, T., Brackey, R., & Liu, J. (2025). The effects of citations and confirmation bias on trust in chatbots. *Proceedings of the Human Factors and Ergonomics Society Annual Meeting.* doi:10.1177/10711813251357884

Craig, C., Mabry, M., Drahos, B., Kessler, W., **Lenz, T.**, Olander, K., & Morris, N. (2024). An assessment of two pedestrian safety inventories. *Advances in Human Factors of Transportation, AHFE International Conference.* doi:10.54941/ahfe1005242

Grant proposal: "Collaborative Research: SCH: VR-Enhanced Digital Twin for Adaptive Nurse Fatigue Management," NSF Smart and Connected Health Program (coauthor, under review), 2026.

Selected Presentations

- "A Survey of Future Teachers Suggests Cautious Openness to VR and Humanoid Robot Use," HFES Annual Meeting, Reno, NV (poster), Fall 2026.
- "A Systematic Review of Human Factors Solutions to Improve Autonomous Vehicle Remote Assistant Situation Awareness," IISE Annual Conference, Arlington, TX (lecture), Spring 2026.
- "The Effects of Citations and Confirmation Bias on Trust in Chatbots," HFES Annual Meeting, Chicago, IL (poster), Fall 2025.
- "Effect of Personalized Driving Algorithms on Trust and Workload," HFES Midwest Regional Meeting, West Lafayette, IN (lightning talk), Spring 2025.
- Invited guest lectures: "Cognitive Ergonomics Sampler" (IE 2710, Spring 2026) and "My Research Journey" (HCI 5790X, Fall 2025), Iowa State University.

Leadership and Mentoring

- **President**, HFES Student Chapter at Iowa State University, Spring 2026 – Present (Vice President, Fall 2025 – Spring 2026).
- **Senator**, Iowa State University Graduate and Professional Student Senate, Fall 2025 – Present.
- Mentored nine undergraduate researchers through NSF REU projects and research assistantships, Spring 2025 – Present.
- Represent the HIA Lab at monthly departmental recruitment events, Spring 2025 – Present.
- Member: Human Factors and Ergonomics Society; Institute of Industrial and Systems Engineers; ISU Human Computer Interaction Student Group.